

Two-Direction Recursive Ritz Refresh

An At-Most-Nine-Dimensional Closure of the Four-Step Schur Packet Chain

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Abstract

RH-92 replaced pointwise sub-quarter decay by a four-step contraction budget, but refreshed the old packet from an ambient leading eigenspace at every update. This paper asks whether the refresh itself can remain recursively low dimensional.

Let V be a rank- r packet and choose k orthonormal directions $Q \subset \text{Ran}(I - VV^*)$. For

$$H = [V, Q]^* G[V, Q] = \begin{pmatrix} A & B \\ B^* & D \end{pmatrix},$$

retaining the leading r Ritz directions gains exactly

$$\Delta_k = \text{tr } D - \sum_{i=1}^k \lambda_i^\uparrow(H).$$

The top- k left singular directions of the projected cross operator $(I - VV^*)GV$ maximize captured cross Frobenius energy. For any full-rank trial frame $W \in \mathbb{C}^{(r+k) \times k}$, Ky Fan's principle gives the reference-free certificate

$$\text{tr}((W^*W)^{-1}W^*HW) + \delta \leq \text{tr } D \implies \Delta_k \geq \delta.$$

This formulation is invariant under a nonorthonormal numerical frame.

We then use the corrected Ritz packet itself as the next packet, with no ambient refresh inside the block. A 384-bit audit compares one, two, and three projected-cross directions over the ten RH-92 windows. The recursive one-direction block has geometric mean above 0.24 in four fine channels, reaching 0.32213305. Two directions reverse all four failures: all forty two-frame forms are strictly negative, every exact rational block budget is below 0.24^4 , and the largest direct block geometric mean is 0.22850222. The top two directions capture at least 96.376% of the projected-cross energy, and the compressed dimension is at most nine. Three directions reduce the worst endpoint tail/reference-tail ratio to 1.05595.

Thus the ambient in-block packet reset is removed from the frozen certificate, and the second direction is genuinely necessary there. No uniform all-level two-direction law, continuum cross-direction construction, Stage-A closure, Hilbert–Polya operator, or Riemann Hypothesis result is claimed.

Keywords: Rayleigh–Ritz; recursive packet; Ky Fan principle; cross Gramian; reduced refresh; validated numerics.

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1 Introduction

The late-memory packet route has progressively reduced an ambient effective-rank question to small variational objects. RH-89 showed that one complement direction gives a strong isolated Ritz correction [8]. RH-90 converted its required gain into a Schur trial sign [10]. RH-91

extracted a conditional tail bootstrap [7], and RH-92 weakened the overly rigid pointwise target to a four-step product budget [6].

One structural gap remained visible in RH-92. At every update, its old packet was recomputed as the leading rank- r packet of the new ambient Gramian. The rank-one Ritz packet certified the local energy gain but was not itself carried into the next update. Consequently the finite certificate still used an ambient spectral refresh between two small Schur solves. This was exactly why the revised Stage-A frontier kept the Schur block gate and the reduced-refresh gate separate.

The present paper tests the most economical way to remove that reset. Start with one leading seed packet at the beginning of the four-step window. At each later update:

1. apply the new Gramian to the current packet and project away the packet;
2. retain the first k left singular directions of this projected cross operator;
3. solve one $(r + k)$ -dimensional Ritz problem;
4. use its leading rank- r packet as the input to the next update.

No ambient eigenspace is inserted inside the chain. The projected cross operator has only r columns, so its nonzero singular data can be found from an $r \times r$ Gram matrix after applying G to the current packet. The remaining spectral solve has dimension $r + k$.

There are three mathematical tasks. First, the rank-one gain identity must be extended to arbitrary complement width k . Second, the complement directions need a canonical optimality statement. Third, the scalar Schur trial must be replaced by a k -frame certificate that remains valid for an almost orthonormal numerical frame. These are supplied in Sections 2 and 3.

The numerical conclusion is unusually sharp. Width one is not merely less accurate: its recursive block target fails in all four channels at $\sigma = 0.02$ and 0.01 . Width two passes all ten channels. Width three is not needed for contraction, but nearly reproduces the endpoint leading packet in every channel. This gives a genuine finite threshold between one and two directions while separating target sufficiency from near-optimal tracking.

2 Multi-direction complement Ritz refresh

Let $G = G^* \geq 0$ act on a finite-dimensional Hilbert space $\mathcal{K}$. Let $V : \mathbb{C}^r \rightarrow \mathcal{K}$ and $Q : \mathbb{C}^k \rightarrow \mathcal{K}$ be isometries with $V^*Q = 0$. Put $Z = [V, Q]$ and partition

$$H = Z^*GZ = \begin{pmatrix} A & B \\ B^* & D \end{pmatrix}, \quad A \in \mathbb{C}^{r \times r}, \quad D \in \mathbb{C}^{k \times k}. \quad (1)$$

Write the eigenvalues of H in increasing order as $\lambda_1^\uparrow(H) \leq \dots \leq \lambda_{r+k}^\uparrow(H)$.

Theorem 2.1 (*k*-direction complement Ritz theorem). *Let C contain the leading r orthonormal eigenvectors of H and put $U = ZC$. Then*

$$\mathrm{tr}(U^*GU) = \mathrm{tr} H - \sum_{i=1}^k \lambda_i^\uparrow(H), \quad (2)$$

$$\Delta_k := \mathrm{tr}(U^*GU) - \mathrm{tr}(V^*GV) = \mathrm{tr} D - \sum_{i=1}^k \lambda_i^\uparrow(H) \geq 0. \quad (3)$$

Consequently

$$E_r(G) \leq \mathcal{E}_G(U) \leq \mathcal{E}_G(V), \quad \mathcal{E}_G(W) = \mathrm{tr} G - \mathrm{tr}(W^*GW) \quad (4)$$

for isometric packets.

Proof. The leading rank- r Ritz capture is the sum of the largest r eigenvalues of H , which equals the right side of (2). Since $\text{tr } H = \text{tr } A + \text{tr } D$ and the old coordinate packet captures $\text{tr } A$, subtraction gives (3). The old packet range is an admissible rank- r subspace of $\text{Ran } Z$, so Ky Fan's maximum principle gives $\Delta_k \geq 0$. The full leading packet can only capture more energy than a Ritz packet restricted to $\text{Ran } Z$ [1, 2]. $\square$

For $P = VV^*$, define the projected cross operator

$$K = (I - P)GV : \mathbb{C}^r \longrightarrow \text{Ran}(I - P). \quad (5)$$

It can be formed without constructing a basis for the full complement: $K = GV - V(V^*GV)$.

Proposition 2.2 (Top- k projected-cross selection). *Let $s_1(K) \geq s_2(K) \geq \dots$ be the singular values of K . Among all isometries $Q : \mathbb{C}^k \rightarrow \text{Ran}(I - P)$,*

$$\max_Q \|Q^*GV\|_{S_2}^2 = \sum_{i=1}^k s_i(K)^2. \quad (6)$$

The maximizing range is spanned by the first k left singular vectors of K , and for this choice

$$\|(I - QQ^*)K\|_{S_2}^2 = \sum_{i>k} s_i(K)^2. \quad (7)$$

Proof. Since $Q^*V = 0$, $Q^*GV = Q^*K$. Ky Fan's maximum principle applied to KK^* proves (6); the singular-value expansion then gives (7) [3]. $\square$

This proposition does not by itself bound the corrected packet tail. It identifies the canonical k -dimensional part of the cross interaction. The complement diagonal block D and the joint Ritz rotation still matter. The audit therefore records cross-energy capture as a mechanism diagnostic, not as a substitute for the final gain certificate.

Corollary 2.3 (Nested enrichment monotonicity). *If the selected complement spaces are nested, $\text{Ran } Q_k \subset \text{Ran } Q_{k+1}$, then the restricted optimal tail is nonincreasing in k at one fixed update.*

Proof. The rank- r trial spaces available in $\text{Ran}[V, Q_k]$ form a nested family. Apply the variational principle. $\square$

The corollary is one-step only. Recursive width- k chains follow different packets after the first update, so their later comparison is not a formal consequence of nesting.

3 Generalized trial-frame gain certificate

The gain in (3) is controlled by the sum of the bottom k eigenvalues of H . A k -dimensional analogue of the RH-90 trial vector is therefore a trial frame for the bottom spectral subspace.

For any full-column-rank $W \in \mathbb{C}^{(r+k) \times k}$, define its generalized Rayleigh trace

$$\mathcal{R}_H(W) = \text{tr}((W^*W)^{-1}W^*HW). \quad (8)$$

This quantity depends only on $\text{Ran } W$, not on the chosen frame coordinates.

Theorem 3.1 (Generalized trial-frame gain certificate). *For every full-rank trial frame W ,*

$$\sum_{i=1}^k \lambda_i^\uparrow(H) \leq \mathcal{R}_H(W). \quad (9)$$

Hence, for any requested gain $\delta \geq 0$,

$$\boxed{\Psi_{\delta,k}(W) := \mathcal{R}_H(W) + \delta - \text{tr } D \leq 0 \implies \Delta_k \geq \delta.} \quad (10)$$

Proof. Let $M = W^*W \succ 0$ and set $Y = WM^{-1/2}$. Then $Y^*Y = I_k$, and cyclicity of the trace gives

$$\text{tr}(Y^*HY) = \text{tr}(M^{-1}W^*HW) = \mathcal{R}_H(W).$$

Ky Fan's minimum principle identifies the sum of the bottom k eigenvalues as the minimum of this trace over all orthonormal k -frames. This proves (9). Combine it with (3). $\square$

The metric factor $(W^*W)^{-1}$ makes the certificate exactly invariant under a nonorthonormal floating frame. No post hoc assumption that the computed eigenvectors are exactly orthogonal is needed. For $k = 2$, the metric inverse is evaluated by its explicit 2×2 determinant in the interval audit.

Theorem 3.2 (Recursive reduced block closure). *Let $G_j \geq 0$ be a sequence of Gramians. Start with an isometric rank- r packet V_b . At update $j > b$, construct k projected-cross directions, form the enriched compression, and let V_j be its leading rank- r Ritz packet. Define*

$$E_{j-1} = \mathcal{E}_{G_{j-1}}(V_{j-1}), \quad (11)$$

$$C_j = \mathcal{E}_{G_j}(V_{j-1}), \quad (12)$$

$$\delta_j = C_j - \rho_j E_{j-1}, \quad \rho_j \geq 0. \quad (13)$$

If either $\delta_j \leq 0$ or a trial frame satisfies $\Psi_{\delta_j, k}(W_j) \leq 0$, then

$$E_j := \mathcal{E}_{G_j}(V_j) \leq \rho_j E_{j-1}. \quad (14)$$

Consequently, over m recursive updates,

$$E_{b+m} \leq \left(\prod_{j=b+1}^{b+m} \rho_j \right) E_b. \quad (15)$$

No ambient packet refresh occurs between the updates.

Proof. If $\delta_j \leq 0$, the old packet already has new tail $C_j \leq \rho_j E_{j-1}$, and Ritz optimization can only improve it. Otherwise Theorem 3.1 gives $\Delta_k \geq \delta_j$. By Theorem 2.1, the corrected tail is $C_j - \Delta_k \leq C_j - \delta_j = \rho_j E_{j-1}$. Iteration proves (15). $\square$

This theorem merges local correction and refresh: the output packet is the next input packet. It still requires one seed packet and a uniform method for constructing the projected-cross directions.

4 384-bit recursive audit

4.1 Models and block protocol

We use the five frozen Gaussian scales $\sigma = 0.16, 0.08, 0.04, 0.02, 0.01$, two directional channels per scale, $\eta = 1/512$, and clock ranks 4, 5, 6, 6, 7. The four-step windows are the same as RH-92: 1–4, 3–6, 8–11, 14–17, and 19–22.

Each chain is seeded once by the leading packet at the time immediately before the block. Thereafter width one, two, or three is propagated recursively. The two-direction chain is the primary certificate. The one-direction chain is a negative comparator, and width three measures how closely a slightly larger packet tracks the endpoint leading reference. The reference packet is not inserted into any recursive chain.

At a primary update, the implementation forms

$$K_j = G_j V_{j-1} - V_{j-1} (V_{j-1}^* G_j V_{j-1}),$$

extracts its first two left singular vectors, and solves an $(r + 2)$ -dimensional Ritz problem. The largest primary compression is 9×9 .

4.2 Exact-binary validation layers

The binary64 pilot selects upward-rounded thousandth budgets, which are then frozen as exact rational numbers. The 384-bit audit checks:

1. the generalized two-frame form in (10);
2. the direct recursive tail ratio for each update;
3. the exact four-step endpoint/seed tail ratio;
4. positivity of the 2×2 frame metric determinant;
5. orthogonality defects of the enriched and corrected packets.

As in RH-92, ambient residuals use the metric-corrected expression

$$\mathcal{E}_G(W) = \text{tr } G - 2 \text{tr}(W^*GW) + \text{tr}((W^*W)(W^*GW)), \quad (16)$$

which is valid for the exactly lifted binary basis. Compressed frame forms and ambient recursive tails are therefore certified separately. This is the same distinction between a small variational model and its floating ambient realization used in validated numerical linear algebra [4, 5].

5 Results: one direction fails, two directions close

σ	side	rank	times	two-direction budgets	one mean	two mean	min cross-2	three/ref
0.16	L	4	1–4	.007/.005/.004/.003	.015121	.004011	.999380	1.000001
0.16	R	4	1–4	.011/.006/.003/.003	.020379	.004200	.999982	1.000006
0.08	L	5	3–6	.012/.006/.007/.017	.037582	.008976	.999989	1.000461
0.08	R	5	3–6	.008/.039/.003/.021	.053466	.010914	.999958	1.007250
0.04	L	6	8–11	.079/.081/.056/.026	.105841	.054339	.999999	1.000220
0.04	R	6	8–11	.036/.065/.180/.089	.091346	.077737	.999999	1.055945
0.02	L	6	14–17	.455/.145/.120/.233	.270599	.206639	.963762	1.000529
0.02	R	6	14–17	.137/.256/.356/.113	.315010	.193248	.995872	1.000578
0.01	L	7	19–22	.335/.248/.168/.197	.316481	.228502	.990854	1.001758
0.01	R	7	19–22	.343/.250/.123/.153	.322133	.199990	.999704	1.005420

Table 1: Recursive packet results. Means are fourth roots of interval block ratios. “Min cross-2” is the smallest top-two projected-cross energy fraction in the channel. “Three/ref” is the endpoint width-three tail divided by the endpoint leading-packet tail.

The one-direction failures are exactly the four fine channels at $\sigma = 0.02$ and 0.01 . Their geometric means are rigorously above 0.24 . The largest is 0.32213305 . Thus the RH-89/RH-90 one-direction correction, although strong when restarted from a leading packet, does not recursively carry its own packet through these four windows.

Two directions close all ten blocks. The hardest rational budget is the left channel at $\sigma = 0.01$:

$$0.335 \cdot 0.248 \cdot 0.168 \cdot 0.197 = 0.00274961568 = 0.82875666 \cdot 0.24^4.$$

Its direct interval geometric mean is at most 0.22850222 . The largest individual two-direction contraction is 0.45462724 , so the success remains a block phenomenon rather than a pointwise sub-quarter law.

The cross-energy diagnostic explains why the second direction matters. At the hardest update the leading cross direction captures only 58.616% of the projected-cross Frobenius energy, whereas the first two capture 96.376% . Across most other updates the top-two fraction is above 99.9% .

validated quantity	worst certified value
generalized two-frame signs	40/40 strict
direct two-direction target contractions	40/40
one-direction failed blocks	4/10
two-direction green blocks	10/10
largest one-direction block mean	0.32213305
largest two-direction budget mean	0.22899079
largest two-direction direct mean	0.22850222
largest three-direction direct mean	0.22620607
minimum top-one cross-energy fraction	0.586156
minimum top-two cross-energy fraction	0.963762
smallest negative two-frame margin	6.2621×10^{-18}
largest primary compressed dimension	9
largest two-direction endpoint/reference ratio	15.916
largest three-direction endpoint/reference ratio	1.05595

Table 2: Global 384-bit summary. The endpoint/reference ratio is a tracking diagnostic, not part of the contraction certificate.

Two directions are sufficient for the target but not uniformly near-optimal. In the coarse right channel at $\sigma = 0.08$, its endpoint tail is about 15.9 times the very small leading-packet tail, even though its four-step contraction is excellent. Width three reduces the worst such ratio over all channels to 1.05595. This distinction is useful: a theorem aimed only at the Stage-A residual may prefer width two, while a theorem that needs stable packet identification may pay for a third direction.

6 Revised frontier and limitations

RH-92 expressed the preferred Stage-A corridor as a block Schur law S_{blk} , a reduced refresh/future gate R , and the inherited prefix/normalization/observability bridge O . The present recursive theorem suggests a more economical combined gate:

$$D_2 : \quad \text{a uniform two-direction projected-cross Ritz chain} \\ \text{with a contracting fixed-length block product.}$$

If D_2 is proved all-level, the sufficient frontier becomes

$$A_1 = L \vee (D_2 \wedge O), \tag{17}$$

where L is the inherited full-block corridor. In this formulation D_2 supplies both the block contraction and the in-block reduced refresh.

The finite result does not prove D_2 . Its remaining components are explicit:

- construct the initial seed packet uniformly at the clock rank;
- control the first two singular directions of $(I - P)GP$ without frozen binary models;
- prove a repeated block product below one after a uniform burn-in;
- propagate the resulting packet through the physical future with the normalization and observability factors in O .

The ambient leading eigensolve is removed only *inside* the frozen four-step block. Applying G to the current rank- r packet remains an ambient operation, though its singular analysis is

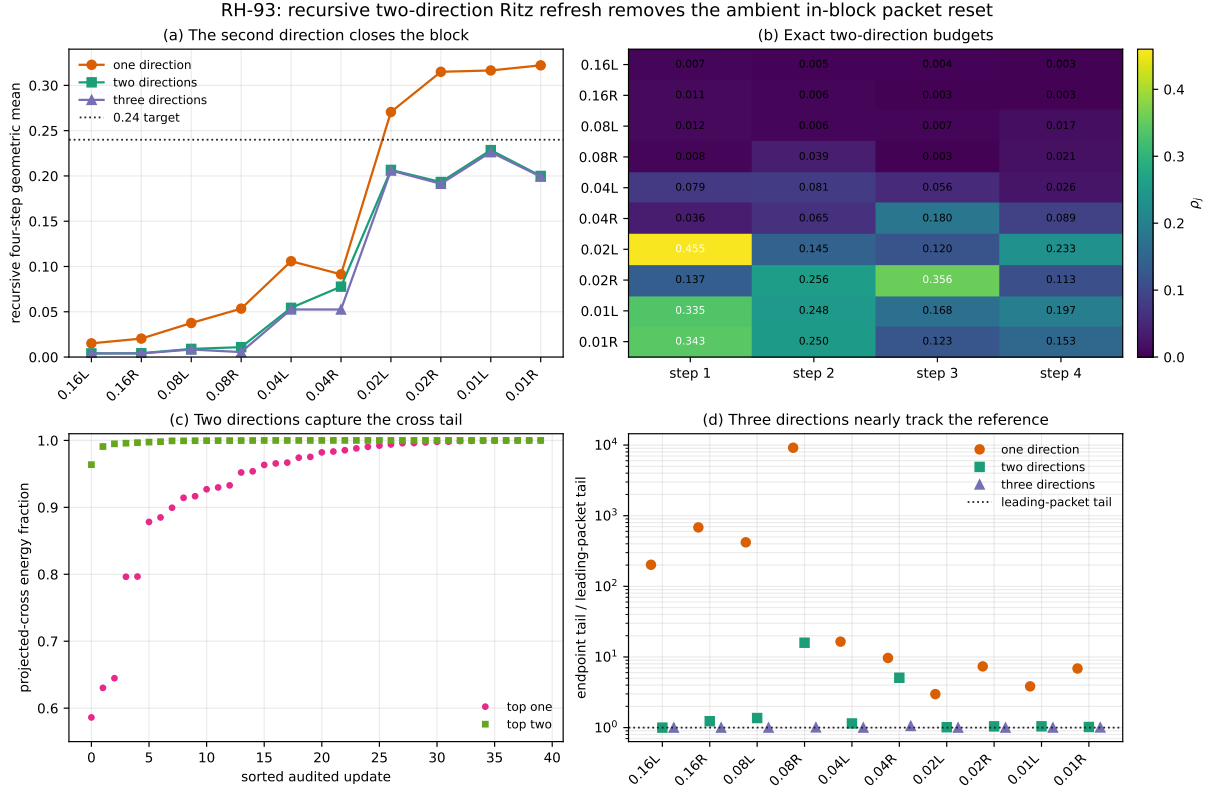


Figure 1: Recursive complement width. Panel (a) shows the four fine failures of width one and universal width-two closure. Panel (b) displays the exact rational two-direction budgets. Panel (c) compares top-one and top-two projected-cross energy capture. Panel (d) shows endpoint tracking relative to the ambient leading packet; width three is uniformly close at the stored anchors.

only r -column and its spectral solves are at most nine dimensional. No complexity theorem or continuum limit for those actions is proved.

The one-direction failure is branch-specific. A different rank staircase, block phase, or complement selector may repair it. Conversely, the three-direction near-reference observation is finite evidence, not a uniform subspace theorem.

The RH-81 moving-cloud projection, coefficient bridge, and uniform trace-class complement are unchanged [9]. This paper constructs no canonical scattering completion, self-adjoint generator, intrinsic $T \log T$ law, prime-power trace formula, completed-zeta identity, Hilbert–Polya operator, zeta-zero identification, or proof of the Riemann Hypothesis.

7 Reproducibility

The archive contains the full and smoke audits, the frozen rational budgets, all forty generalized frame forms, exact-binary recursive residuals for widths one through three, cross singular-value diagnostics, generated figures, unit tests, and SHA-256 manifests for local sources, dependencies, and publication artifacts. The accompanying README lists the reproduction commands. Every interval sign and tail ratio is evaluated after exact binary lifting at 384-bit precision.

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